



Reading Assignment 3

Report

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Ebb-and-Flow Protocols: A Resolution of the Availability-Finality Dilemma

1 Initial Challenges

- The two-ledger model felt conceptually elegant but mechanically unclear. I understood that `LOG_fin` should be a prefix of `LOG_da`, but did not immediately understand how this invariant is preserved under adversarial behavior.
- Adversary models and assumptions were easy to confuse. The paper separates assumptions for finality (partially synchronous, threshold $\sim 1/3$) and dynamic availability (synchronous, threshold $\sim 1/2$). I initially struggled to map which guarantee belongs to which model.
- The balancing attack details on Gasper were hard to simulate mentally. Terms such as committee assignment, swayers, withheld votes, and fork-choice flip-flopping required multiple rereads.
- The theorem statement was concise but dense. I needed extra clarification on why the result is described as “optimal resilience,” and how this relates to known impossibility boundaries.

2 Process of Inquiry

Interaction 1: Understanding the balancing attack on Gasper

My Prompt: “Explain the balancing attack on Gasper in plain language, but preserve technical correctness (committees, withheld votes, and fork-choice dynamics).” “Why can this attack hurt both liveness (no finalization) and eventually safety at the underlying fork-choice layer?”

My Analysis: The response was useful because it separated the attack into a step-by-step timeline: attacker waits for an opportune epoch, creates symmetric pressure on two forks, selectively releases withheld votes to keep honest validators split, prevents $2/3$ support from converging on one chain for finality. This helped me understand that the attack is not merely “equivocation,” but a control of information timing that repeatedly destabilizes convergence.

Interaction 2: Formal meaning of P1, P2, and prefix in Definition 3

My Prompt: “Translate Definition 3 into practical guarantees a wallet user or validator should care about.” “What assumptions switch between P1 and P2, and why is that switch not a contradiction?”

My Analysis: The strongest part of the response was framing the protocol as intentionally flexible by client risk preference: conservative clients rely on `LOG_fin` (partition-safe finality under partial synchrony assumptions), aggressive clients use `LOG_da` (better progress under dynamic availability assumptions). The assistant correctly emphasized that these are not simultaneous guarantees under one identical environment; they are guarantees under different

modeled conditions. This removed my earlier misconception that the paper was “violating CAP.”

Interaction 3: Why snap-and-chat needs boycott logic

My Prompt: “If we already run a BFT protocol on snapshots, why do we need additional boycott checks?” “Give an attack sketch if boycott is removed.”

My Analysis: This was one of the most valuable exchanges. The assistant explained that without boycott, an adversary can propose a snapshot that appears structurally valid to BFT but contains unconfirmed content from the dynamically available layer. If BFT finalizes that snapshot, LOG_{fin} could include state that should not be trusted, breaking intended safety semantics and potentially violating the intended linkage with LOG_{da}.

3 Synthesis and Lingering Questions

My current understanding is that the paper makes three core moves:

Diagnose a fundamental trade-off (availability vs. finality under dynamic participation and partitions).

Critique a major Ethereum PoS direction (Gasper) by showing a balancing attack that prevents finalization and stresses security assumptions.

Provide a compositional alternative (snap-and-chat) where: a dynamically available protocol maintains a growing available chain, snapshots are fed to a BFT protocol for finalized prefix, boycott filtering and ledger extraction preserve prefix consistency.